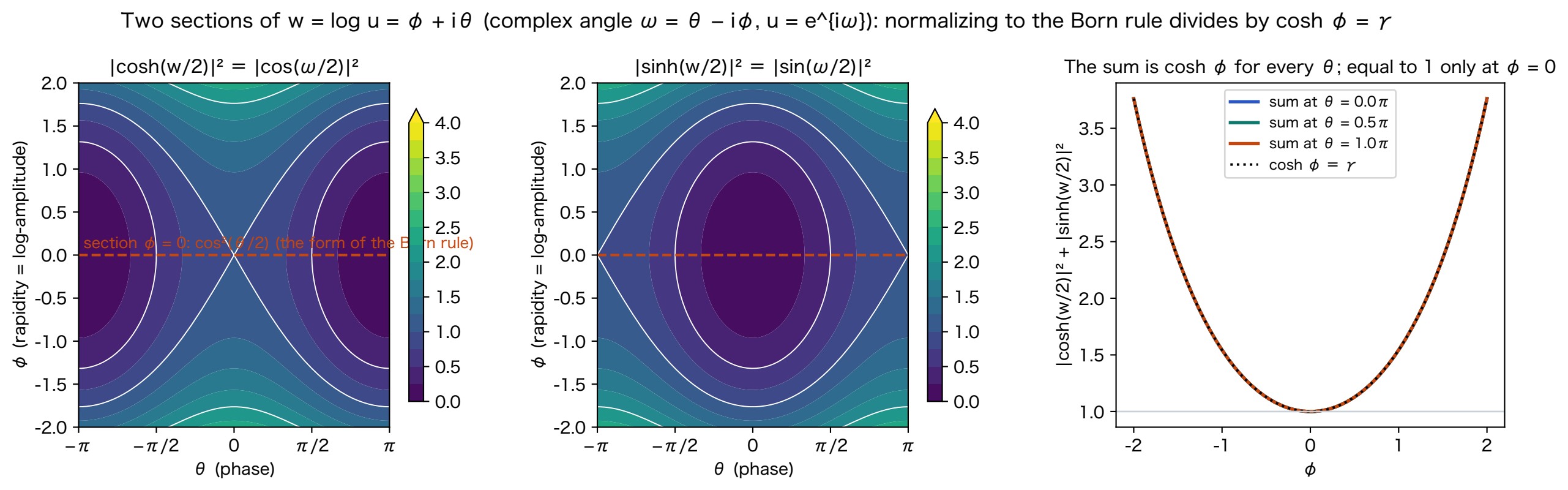

Two sections of w = log u = ϕ + iθ (complex angle ω = θ − iϕ, u = e^{iω}): normalizing to the Born rule divides by cosh ϕ = r
|cosh(w/2)|² = |cos(ω/2)|²
|sinh(w/2)|² = |sin(ω/2)|²
The sum is cosh ϕ for every θ; equal to 1 only at ϕ = 0
section ϕ = 0: cos²(θ/2) (the form of the Born rule)
ϕ (rapidity = log-amplitude)
θ (phase)
−π
−π/2
0
π/2
π
2.0
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
|cosh(w/2)|² + |sinh(w/2)|²
ϕ
-2
-1
0
1
2
3.5
3.0
2.5
2.0
1.5
1.0
sum at θ = 0.0π
sum at θ = 0.5π
sum at θ = 1.0π
cosh ϕ = r